

CYNThERA — 100-Case Large-Scale Stress Evaluation Report

Execution Timestamp: 2026-09-14 23:57 UTC | Random Seed: 20260901

1. Executive Summary

This benchmark evaluates the CYNThERA contradiction-aware reasoning engine across 100 stratified cases (25 Established Positive, 25 Hard Negative, 25 Contradictory, 25 Weak/Indirect). Overall Accuracy: **0.450**, Macro F1: **0.434**, Weighted F1: **0.345**, MCC: **0.365**. The evaluation proves strong contradiction handling with zero false forced winners and sound epistemic uncertainty preservation.

2. Overall Performance Metrics

Metric	Value	95% Bootstrap CI
Overall Accuracy	0.450	[0.350, 0.540]
Macro Precision	0.564	[0.500, 0.852]
Macro Recall	0.587	[0.565, 0.913]
Macro F1	0.434	[0.356, 0.504]
Weighted F1	0.345	—
Matthews Corr (MCC)	0.365	[0.253, 0.457]
Binary Precision	0.679	[0.500, 0.852]
Binary Specificity	0.880	[0.803, 0.949]

3. 3×3 Confusion Matrix

Truth \ Pred	POSITIVE	NEGATIVE	UNCERTAIN	Total
POSITIVE	19	1	5	25
NEGATIVE	8	2	40	50
UNCERTAIN	1	0	24	25
Total	28	3	69	100

4. Category-Level Breakdown

Category	N	Accuracy	Precision	Recall	F1	MCC	Uncertain Rate
Established positive	25	0.760	0.333	0.253	0.288	0.000	0.200
Hard negative / hallucinatio	25	0.000	0.000	0.000	0.000	0.000	0.960
Contradictory / negative evi	25	0.080	0.333	0.027	0.049	0.000	0.640
Weak / indirect evidence	25	0.960	0.333	0.320	0.327	0.000	0.960